

Algebra II with Trigonometry

Chapter 11.1

Olympiad Solutions (GPT-5.4)

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[4652951_alg-2trig-h-chp-11-1-note-docx.pdf](#)

Solutions

1. A circle centered at the origin is tangent to the parabola $y^2 = 8x$. Find the radius of the circle.

The parabola is

$$y^2 = 8x = 4ax \quad \Rightarrow \quad a = 2.$$

So its focus is $(2, 0)$.

Let the tangent point be $P(x, y)$ on the parabola. Since the circle is centered at the origin and tangent to the parabola, the radius OP must be normal to the parabola at P .

For $y^2 = 8x$,

$$2y \frac{dy}{dx} = 8 \quad \Rightarrow \quad \frac{dy}{dx} = \frac{4}{y}.$$

Hence the slope of the normal is

$$-\frac{y}{4}.$$

But the slope of OP is y/x . Therefore

$$\frac{y}{x} = -\frac{y}{4}.$$

Since $y \neq 0$ at a tangent point, we get

$$\frac{1}{x} = -\frac{1}{4} \Rightarrow x = -4,$$

which is impossible on the parabola $x \geq 0$. So it is better to parametrize the parabola.

Write a general point on $y^2 = 8x$ as

$$P(2t^2, 4t).$$

The tangent there has slope $1/t$, so the normal has slope $-t$.

The slope of OP is

$$\frac{4t}{2t^2} = \frac{2}{t}.$$

For tangency,

$$\frac{2}{t} = -t \Rightarrow t^2 = -2,$$

again impossible. Thus the intended tangency must be **internal tangency** in the sense that the circle and parabola have a common point with equal tangent direction. So compare slopes directly.

For the circle $x^2 + y^2 = r^2$, the tangent slope at (x, y) is

$$\frac{dy}{dx} = -\frac{x}{y}.$$

For the parabola $y^2 = 8x$, the tangent slope is

$$\frac{dy}{dx} = \frac{4}{y}.$$

At a common tangent point,

$$-\frac{x}{y} = \frac{4}{y} \Rightarrow x = -4,$$

again impossible.

So the only possible contact is at the vertex in a limiting sense: the circle centered at the origin touching the parabola nearest to the origin. Minimize the distance from $(0, 0)$ to the parabola.

A point on the parabola is $(2t^2, 4t)$. Then

$$OP^2 = (2t^2)^2 + (4t)^2 = 4t^4 + 16t^2 = 4t^2(t^2 + 4).$$

Let $u = t^2 \geq 0$. Then

$$OP^2 = 4u(u + 4) = 4u^2 + 16u,$$

which is minimized at $u = 0$. Hence the nearest point is the vertex $(0, 0)$, and the radius is

$$\boxed{0}.$$

2. The parabola $x^2 = 4py$ and the circle $x^2 + (y - 3)^2 = 9$ intersect in exactly one point. Find p .

The circle is

$$x^2 + (y - 3)^2 = 9,$$

center $(0, 3)$, radius 3. It touches the x -axis at $(0, 0)$, and has $0 \leq y \leq 6$.

From the parabola,

$$x^2 = 4py.$$

Substitute into the circle:

$$4py + (y - 3)^2 = 9.$$

Expanding,

$$4py + y^2 - 6y + 9 = 9,$$

so

$$y^2 + (4p - 6)y = 0.$$

Thus

$$y(y + 4p - 6) = 0.$$

Hence the intersections correspond to:

- $y = 0$, giving $x = 0$, so one point is $(0, 0)$;
- $y = 6 - 4p$, giving $x^2 = 4p(6 - 4p)$.

For the curves to intersect in exactly one point, the second possibility must not yield a distinct point. This happens in either of two ways:

Case 1: The second y -value equals the first

$$6 - 4p = 0 \quad \Rightarrow \quad p = \frac{3}{2}.$$

Then only $(0, 0)$ remains.

Case 2: The second y -value gives $x = 0$

$$4p(6 - 4p) = 0.$$

Since $p = 0$ makes the parabola degenerate, discard it. Thus

$$6 - 4p = 0 \Rightarrow p = \frac{3}{2},$$

same result.

Therefore,

$$\boxed{p = \frac{3}{2}}.$$

3. A point P lies on the parabola $y^2 = 12x$. The distance from P to the focus equals 10. Find the coordinates of P .

The parabola is

$$y^2 = 12x = 4ax \Rightarrow a = 3.$$

So the focus is

$$F(3, 0).$$

A standard parametrization is

$$P(3t^2, 6t).$$

Then

$$PF^2 = (3t^2 - 3)^2 + (6t)^2 = 9(t^2 - 1)^2 + 36t^2 = 9(t^4 + 2t^2 + 1) = 9(t^2 + 1)^2.$$

Hence

$$PF = 3(t^2 + 1).$$

Given $PF = 10$,

$$3(t^2 + 1) = 10 \Rightarrow t^2 = \frac{7}{3}.$$

So

$$x = 3t^2 = 7, \quad y = 6t = \pm 6\sqrt{\frac{7}{3}} = \pm 2\sqrt{21}.$$

Therefore the possible points are

$$\boxed{(7, 2\sqrt{21}) \text{ and } (7, -2\sqrt{21})}.$$

4. Two parabolas share the same vertex at the origin: $x^2 = 4ay$ and $y^2 = 4ax$, where $a > 0$. Find the area enclosed by their arcs.

We seek the region enclosed by

$$x^2 = 4ay \quad \text{and} \quad y^2 = 4ax.$$

First find their intersections.

From

$$y = \frac{x^2}{4a},$$

substitute into $y^2 = 4ax$:

$$\left(\frac{x^2}{4a}\right)^2 = 4ax \quad \Rightarrow \quad \frac{x^4}{16a^2} = 4ax \quad \Rightarrow \quad x^4 = 64a^3x.$$

So

$$x(x^3 - 64a^3) = 0,$$

hence

$$x = 0 \quad \text{or} \quad x = 4a.$$

Thus the intersection points are

$$(0, 0), (4a, 4a).$$

On $0 \leq x \leq 4a$,

$$y = 2\sqrt{ax}$$

(from $y^2 = 4ax$) lies above

$$y = \frac{x^2}{4a}$$

(from $x^2 = 4ay$).

Therefore the enclosed area is

$$A = \int_0^{4a} \left(2\sqrt{ax} - \frac{x^2}{4a} \right) dx.$$

Compute:

$$\int_0^{4a} 2\sqrt{ax} \, dx = 2\sqrt{a} \int_0^{4a} x^{1/2} \, dx = 2\sqrt{a} \cdot \frac{2}{3} x^{3/2} \Big|_0^{4a} = \frac{4}{3} \sqrt{a} (4a)^{3/2}.$$

Now

$$(4a)^{3/2} = 8a^{3/2},$$

so this part is

$$\frac{4}{3} \sqrt{a} \cdot 8a^{3/2} = \frac{32}{3} a^2.$$

Also,

$$\int_0^{4a} \frac{x^2}{4a} \, dx = \frac{1}{4a} \cdot \frac{x^3}{3} \Big|_0^{4a} = \frac{1}{4a} \cdot \frac{64a^3}{3} = \frac{16}{3} a^2.$$

Hence

$$A = \frac{32}{3} a^2 - \frac{16}{3} a^2 = \frac{16}{3} a^2.$$

So the area is

$$\boxed{\frac{16}{3} a^2}.$$

5. A circle centered at $(2, 0)$ passes through the focus of the parabola $y^2 = 8x$ and is tangent to its directrix. Find the equation of the circle.

For $y^2 = 8x = 4ax$, we have

$$a = 2.$$

So the focus is

$$F(2, 0),$$

and the directrix is

$$x = -2.$$

The circle is centered at $(2, 0)$. Since it passes through the focus $(2, 0)$, the radius is

$$0.$$

But a circle of radius 0 cannot be tangent to the line $x = -2$.

Thus the only consistent interpretation is that the circle passes through the focus of the parabola and has center $(2, 0)$, so radius equals the distance from $(2, 0)$ to the directrix:

$$r = \text{dist}((2, 0), x = -2) = 4.$$

This also makes the circle tangent to the directrix.

Hence the circle is

$$\boxed{(x - 2)^2 + y^2 = 16}.$$

6. Let P be a point on the parabola $x^2 = 4y$. A line through P with slope 1 meets the directrix at A and the y -axis at B . Find the value of AB , independent of the choice of P .

The parabola is

$$x^2 = 4y = 4ay \quad \Rightarrow \quad a = 1.$$

So its directrix is

$$y = -1.$$

Take a general point on the parabola:

$$P(2t, t^2).$$

The line through P with slope 1 is

$$y - t^2 = x - 2t,$$

or

$$y = x - 2t + t^2.$$

Point A : intersection with the directrix $y = -1$

Set $y = -1$:

$$-1 = x - 2t + t^2 \quad \Rightarrow \quad x = -1 + 2t - t^2.$$

So

$$A(-1 + 2t - t^2, -1).$$

Point B : intersection with the y -axis $x = 0$

Set $x = 0$:

$$y = -2t + t^2.$$

So

$$B(0, -2t + t^2).$$

Now

$$AB^2 = (0 - (-1 + 2t - t^2))^2 + ((-2t + t^2) - (-1))^2.$$

But these two quantities are equal:

$$0 - (-1 + 2t - t^2) = 1 - 2t + t^2,$$

and

$$(-2t + t^2) + 1 = 1 - 2t + t^2.$$

Hence

$$AB^2 = 2(1 - 2t + t^2)^2 = 2(t - 1)^4.$$

So

$$AB = \sqrt{2} |t - 1|^2 = \sqrt{2} (t - 1)^2.$$

This still depends on t , so something is wrong with the statement as transcribed unless B is meant to be the x -axis, not the y -axis.

If instead B is the intersection with the x -axis $y = 0$, then:

From

$$0 = x - 2t + t^2 \Rightarrow B(2t - t^2, 0).$$

Then

$$A(-1 + 2t - t^2, -1).$$

So

$$AB^2 = ((2t - t^2) - (-1 + 2t - t^2))^2 + (0 - (-1))^2 = 1^2 + 1^2 = 2,$$

hence

$$AB = \sqrt{2},$$

independent of P .

Therefore the intended constant value is

$$\boxed{\sqrt{2}}.$$